

Gyro Gyroscope

A gyroscope is a device for measuring or maintaining orientation, based on the principles of the conservation of angular momentum.

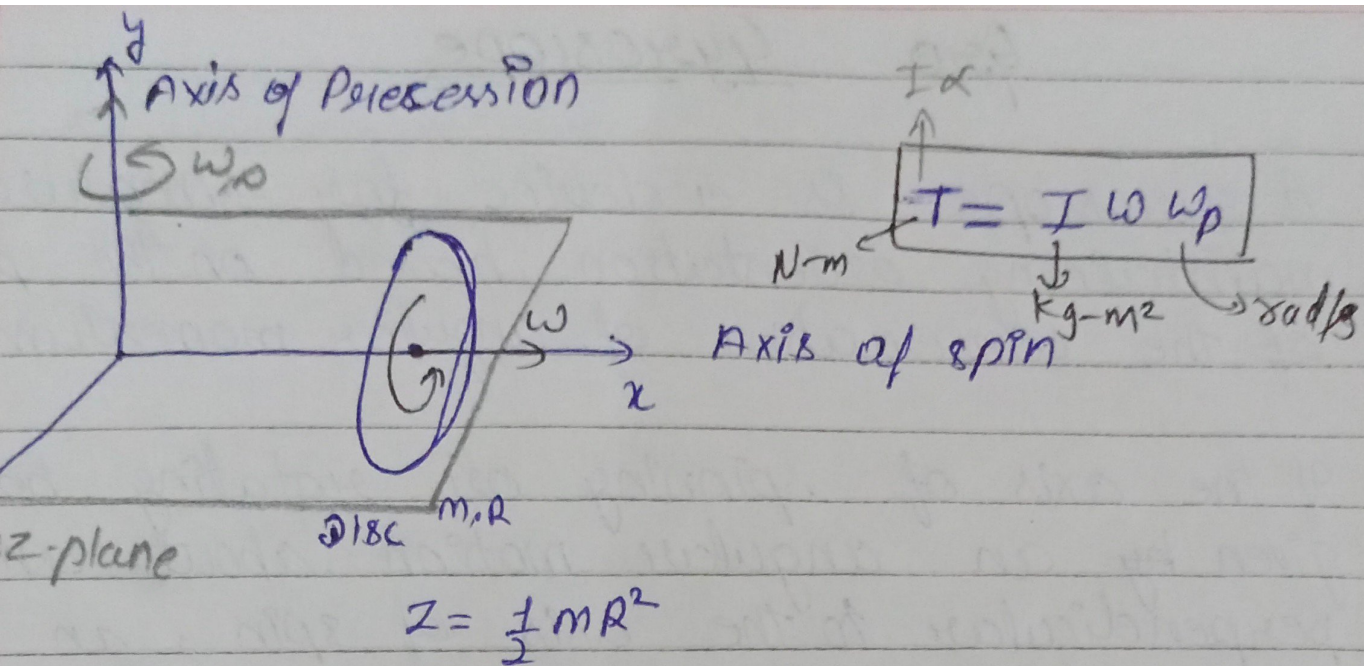
If the axis of spinning or rotating body is given by an angular motion about an axis perpendicular to the axis of spin, an angular acceleration acts on the body about the third perpendicular axis.

The torque required to produce this accelⁿ is known as the active gyroscopic torque.

A reactive gyroscopic torque or couple also acts similar to the concept of centripetal and centrifugal forces on a rotating body.

The effect produced by the reactive gyroscopic couple is known as the gyroscopic effect.

Aeroplane, ships, automobiles, etc, that have rotating parts in the form of wheels or rotors of engines experiences gyroscopic effect while taking turn, such as when axes of spin is subjected to some angular motions.



topic
ple

Axis of spin (ω)	} mutually ⊥ to each other
Axis of precession (ω_p)	
Axis of gyroscopic couple (T)	

Angular velocity

The angular velocity of a rotating body is specified by

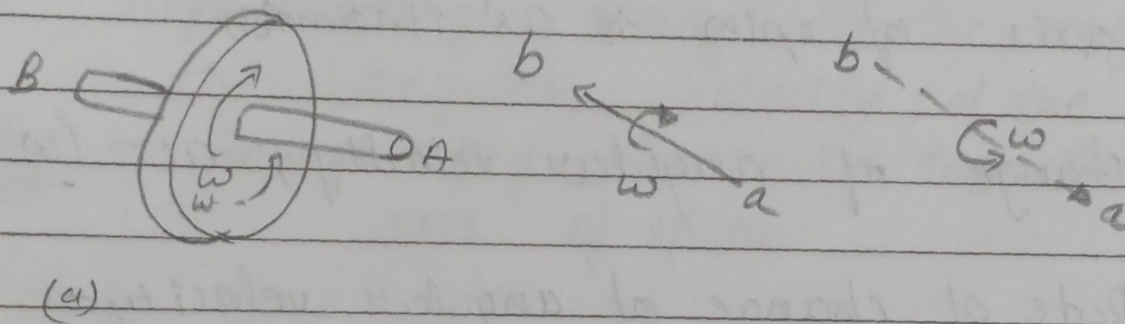
- the magnitude of velocity
- the direction of the axis of rotation.
- the sense of rotation of the rotor, i.e., clockwise or counterclockwise.

Angular velocity is represented by a vector in the following manner.

Direction of Axis of the rotor is represented by drawing the vector parallel to the Axis of the rotor or normal to the plane of the angular velocity.

Sense of rotation of the rotor is denoted by taking the direction of the vector in a set rule. The general rule is that of a right-handed screw.

If a screw is rotational rotated in the clockwise direction, it goes away from the viewer and vice versa.



Shows a rotor which rotates in the clockwise direction when viewed from the end A. The sense of direction of the vector is from a to b according to the screw rule. If the direction of the rotor is reversed, it would be from b to a.

Angular Acceleration -

Let a rotar spin (rotate) about the horizontal axis Ox at a speed of ω rad/s in the direction.

Let (a) represent its angular velocity fig (b)

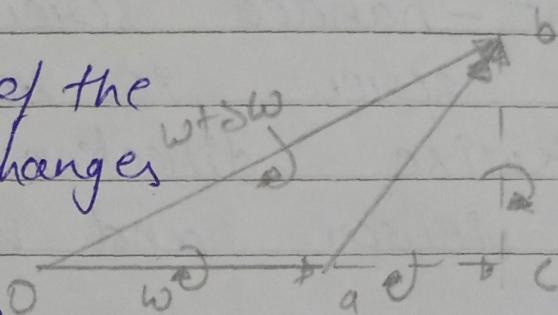
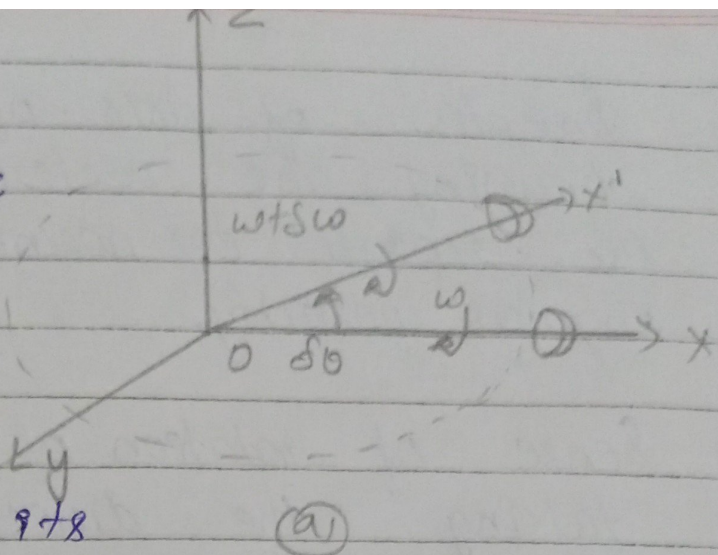
If the magnitude of the angular velocity changes to $(\omega + \delta\omega)$ and the direction of the axis of spin to Ox' (in time δt)

change of angular velocity; $ac = \frac{(\omega + \delta\omega) \cos \delta\theta - \omega}{\delta t}$

Rate of change of angular velocity =
$$\frac{(\omega + \delta\omega) \cos \delta\theta - \omega}{\delta t}$$

$\therefore$ angular Accⁿ =
$$\lim_{\delta t \rightarrow 0} \frac{(\omega + \delta\omega) \cos \delta\theta - \omega}{\delta t}$$

As $\delta t \rightarrow 0$, $\delta\theta \rightarrow 0$ and $\cos \delta\theta \rightarrow 1$



Total angular accelⁿ,

$$\alpha = \frac{d\omega}{dt} \neq \omega \frac{d\theta}{dt}$$

$\frac{d\omega}{dt}$, representing change in the magnitude of the angular velocity of the rotor.
 $\omega \frac{d\theta}{dt}$, representing change in the direcⁿ of the axis of spin.

Gyroscopic torque (couple)

As we know that any rotating body will try not to disturb its axis.

If any external force is applied on the body which tries to change its axis of rotation then a resistive force is generated.

This resistive force is known as gyroscopic couple.

ab $\rightarrow$ Applied couple
b'a' $\rightarrow$ Reaction couple

angular momentum of rotor = $I\omega$

change in angular momentum = $I\omega d\theta$

Rate of change / angular accⁿ = $\frac{I\omega d\theta}{dt}$

Torque / couple = $C = \frac{I\omega d\theta}{dt}$

Gyroscopic couple = $C = I\omega \cdot \omega_p$

The rotor spins in one plane, the axis of rotor precesses in another plane and the gyroscopic couple acts in third plane.

Where - OX - Axis of spin

OZ - axis of precession

OY - axis of gyroscopic couple.

The effect of the gyroscopic couple on a rotating body is known as the gyroscopic effect on the body. A gyroscope is a spinning body which is free to move in other directions under the action of external forces.

A uniform disc having a mass of 8 kg and a radius of gyration of 150 mm is mounted on one end of a horizontal arm of 200 mm length. The other end of the arm can rotate freely in a universal bearing. The disc is given a clockwise spin of 250 rpm as seen from the disc end of the arm. Determine the motion of the disc if the arm remains horizontal.

$$m = 8 \text{ kg} \quad l = 0.2 \text{ m} \quad k = 0.15 \text{ m} \quad N = 240 \text{ rpm}$$

$$I = mk^2 = 8 \times (0.15)^2 = 0.18 \text{ kg.m}^2$$

$$\omega = \frac{2\pi \times 240}{60} = 25.13 \text{ rad/s}$$

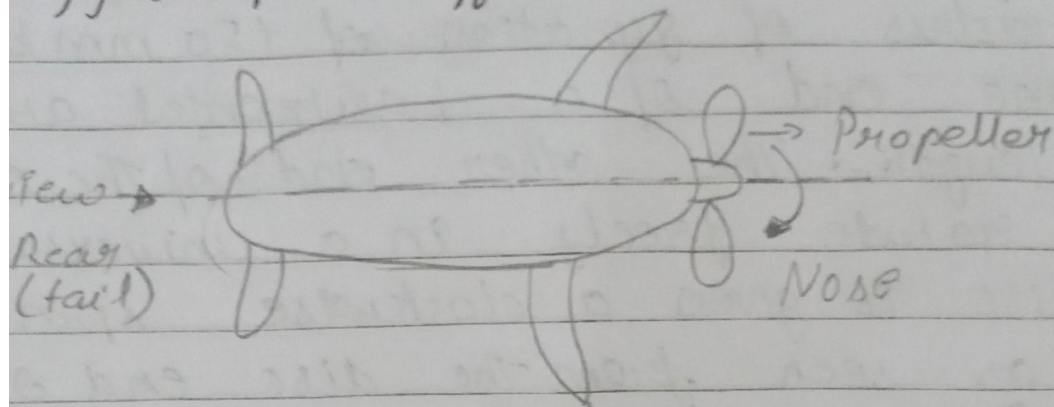
$$C = I\omega\omega_p$$

$$mgl = I\omega\omega_p$$

$$8 \times 9.81 \times 0.2 = 0.18 \times 25.13 \times \omega_p$$

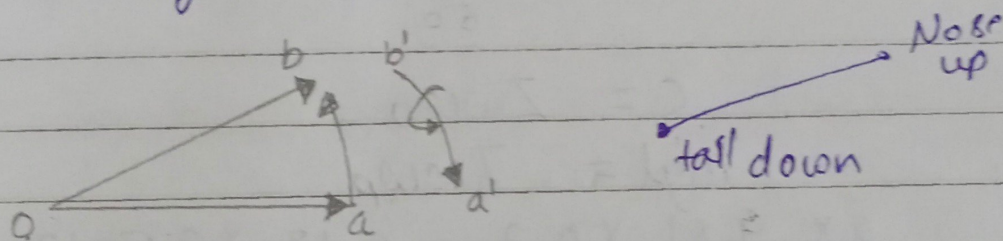
$$\omega_p = 3.47 \text{ rad/s}$$

Gyroscopic effect on Aeroplanes -



Let the propeller be rotating in the clockwise direction when viewed from the rear end.

When turning left



Before taking left turn the angular momentum vector is represented by OA .

When it takes left turn the active gyroscopic couple will change direction of angular momentum from OA to OA' .

The vector AA' is active gyroscopic couple and is $\perp$ to OA .

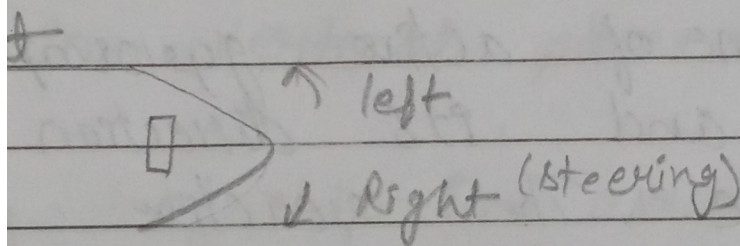
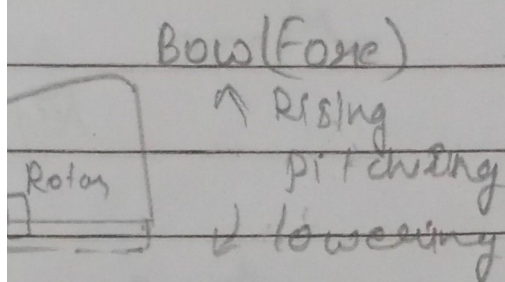
The reactive vector is given by $A'A$ which is equal and opposite to vector AA' .

By applying right hand screw rule we find that the reactive vector is anti clockwise.

∴ Rise the nose and depress the tail.

coptane takes a right turn
plan condition, the reactive
couple will dip the nose and
pl.

effect on Naval Ships $\Rightarrow$



on board

plane of spin of rotor be horizontal,
angular velocity of rotor in
direction when viewed from stern
d).

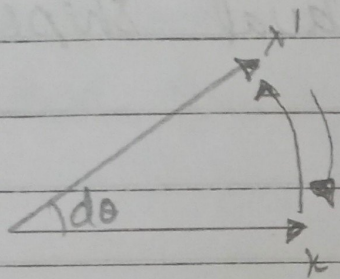
gyroscopic couple to occur, the axis of
should always be $\perp$ to axis of spin.
the axis of precession becomes

parallel to axis of spin as in case of rolling, there will be no gyroscopic couple acting on the ship.

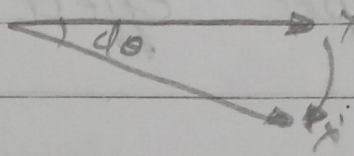
Rolling is limited angular motion of the ship about longitudinal axis.

steering / Gyroscopic effect on Pitching

Steering is the turning of complete ship in a curve towards left / Right.



left



Right

The plane of active gyroscopic couple is $\perp$ to xx' and its direction for left turn is clockwise. The reactive gyroscopic couple of same magnitude will act in opposite direction i.e. in anticlockwise.

The effect of reactive gyroscopic is to raise the bow and lower the stern.

(clockwise + viewing from rear)

Left - up $\rightarrow$ bow (port)

down $\rightarrow$ stern (starboard)

Right $\rightarrow$ up - Stern (starboard)
down - Bow (port)

$$\theta = \phi \sin \omega_0 t$$

ϕ = amplitude

$$\omega_0 = \text{angular velocity of SHM} = \frac{2\pi}{\text{Time period}}$$

$$\frac{d\theta}{dt} = \phi \omega_0 \cos \omega_0 t$$

This is maximum when $\cos \omega_0 t = 1$
maximum angular velocity of precession

$$\omega_p = \phi \omega_0$$

$$\text{Gyroscopic couple } I \omega \omega_p = I \omega \left(\phi \times \frac{2\pi}{T} \right)$$

$$\text{Angular accel}^n = \phi \omega_0^2 \sin \omega_0 t$$

$$\text{Max. ang. accel}^n = \phi \omega_0^2$$

Pitching-

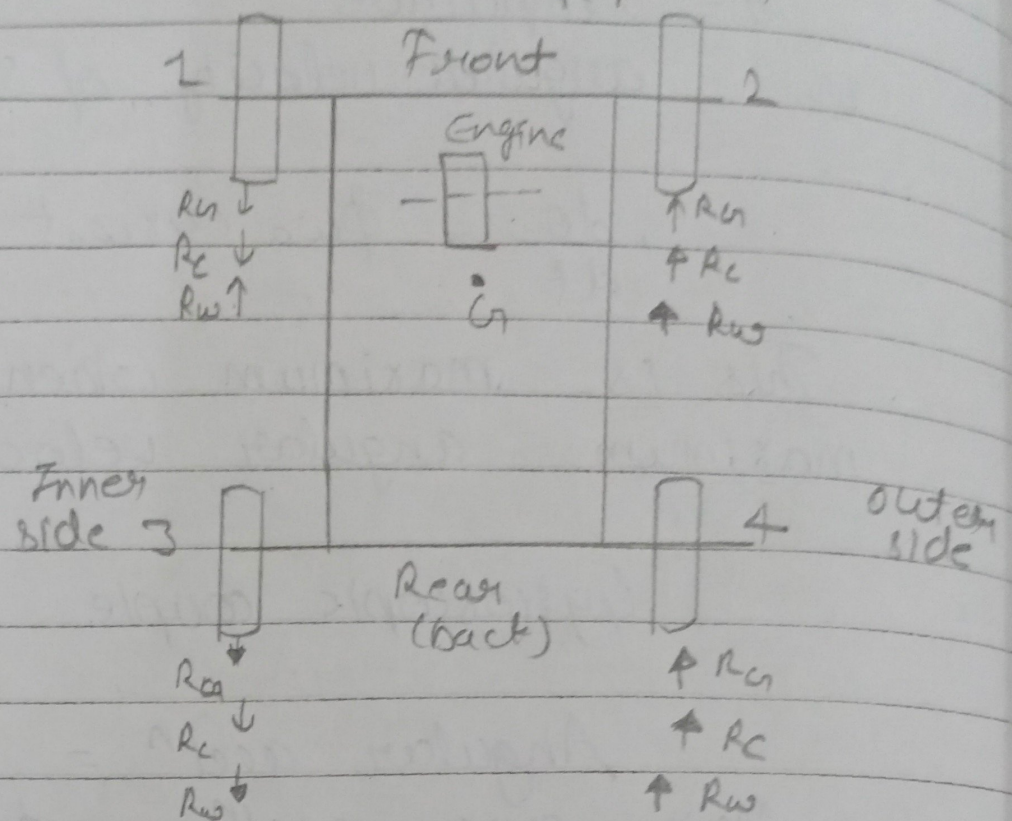
Pitching is the movement of complete ship up down in vertical plane about transverse axis.

The pitching of ship is assumed to take place with simple harmonic motion.

When the pitching is upward, the effect of reactive gyroscopic couple will try to move ship towards star-board.

If pitching is downward the effect of reactive gyroscopic couple is to turn toward port side.

Stability of an Automobile - (left turn)



Reaction due to weight

$$W = mg$$

weight on each wheel = $\frac{W}{4} = \frac{mg}{4} \downarrow$

$R_w =$ Reaction on each wheel = $\frac{W}{4} = \frac{mg}{4} \uparrow$

Reaction due to gyroscopic couple

Gyroscopic couple due to 4 wheel

$$= 4 I_w \omega_w \omega_p = C \omega$$

$$= 4 I_w \frac{V^2}{R}$$

$\omega_w = \frac{V}{R}$; $\omega_p = \frac{V}{R}$

Gyroscopic couple due to engine rotating parts =

$$C_e = I_e \omega_e \omega_p$$

$$= I_e G \omega_a \omega_p$$

$$G = \text{gear ratio} = \frac{\omega_e}{\omega_a}$$

Total Gyroscopic Couple = $C_G = G C_e$
 [positive sign is used when engine parts rotate in same direction as that of wheels]

→ forces on two outer wheels = $\frac{C_G}{2\omega} (\downarrow)$

$$\therefore \text{each wheel} = \frac{C_G}{2\omega} (\downarrow)$$

Forces on two inner wheels (1,3) = $\frac{C_G}{2\omega} (\downarrow)$

→ Reaction on ground on each outer wheel
 $R_G = \frac{C_G}{2\omega} (\uparrow)$

→ Reaction due to centrifugal couple →

$$\text{Centrifugal force} = MR\omega_p^2 = MR\left(\frac{V}{R}\right)^2 = \frac{mV^2}{R}$$

$$\text{Overturning couple} = C_c = MR\omega_p^2 \times h = \frac{mV^2 h}{R}$$

• Forces on two outer wheels (2,4) = $\frac{C_c}{\omega} (\downarrow)$

• Reaction on ground on each outer wheel =
 $R_G = \frac{C_c}{2\omega} (\uparrow)$

$$F_{g21} \text{ positive direction} \rightarrow \frac{\omega}{4} \times \frac{b + c}{2\omega}$$

$$0.21 \quad R_{10} \geq R_{G1} + R_c$$

17.1)
tan

$$201^m \Rightarrow m = 8 \text{ kg}$$

$$l = 0.2 \text{ m}$$

$$k = 0.15 \text{ m}$$

$$N = 240 \text{ rpm}$$

$$I = mk^2$$

$$= 8 \times (0.15)^2 = 0.18 \text{ kg m}^2$$

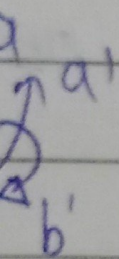
$$\omega = \frac{2\pi \times 240}{60} = 25.13 \text{ rad/s}$$

$$C = I\omega\omega_p$$

$$mgl = I\omega\omega_p$$

$$8 \times 9.81 \times 0.2 = 0.18 \times 25.13 \times \omega_p$$

$$\omega_p = 3.47 \text{ rad/s}$$



Force at bearing B due to gyro couple = 79 N (downwards)
each bearing is to be in the opposite

$$\text{each bearing due to weight of the disc} \\ = \frac{4 \times 9.81}{2} = 19.6\text{ N}$$

each bearing due to weight = 19.6 N (upwards)

$$79 - 19.6 = 59.4\text{ N (upwards)}$$

$$79 + 19.6 = 98.6\text{ N (downwards)}$$

$$R = 50\text{m} \quad v = 200\text{km/hr} = 55.6\text{m/s}$$

$$m = 400\text{kg}, \quad k = 0.3\text{m} \quad N = 2400\text{rpm}$$

$$\omega = 2\pi \times 2400 / 60 = 251\text{rad/s}$$

$$I = mk^2 = 400(0.3)^2 = 36\text{kg-m}^2$$

$$\omega_p = \frac{v}{R} = \frac{55.6}{50} = 1.11\text{rad/s}$$

$$C = I\omega\omega_p = 36 \times 251.4 \times 1.11$$

$$= 10046\text{N-m}$$

$$= 10.046\text{KN-m}$$

When the aeroplane turns towards left, the effect of the gyroscopic couple is to lift the nose upwards and tail downwards.

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$$\Rightarrow m = 450\text{kg}; \quad k = 0.32\text{m}$$

$$\omega = \frac{2\pi \times 2000}{60} = 209.4\text{rad/s}$$

$$v = \frac{240 \times 10^3}{3600} = 66.67\text{m/s}$$

$$I = mk^2 = 450 \times (0.32)^2 = 46.08\text{kg-m}^2$$

$$\omega_p = \frac{v}{R} = \frac{66.67}{60} = 1.11\text{rad/s}$$

$$C = I\omega\omega_p = 46.08 \times 209.4 \times 1.11$$

$$= 10713\text{N-m} = 10.713\text{KN-m}$$

This couple acts in the vertical plane and tends to raise the nose and depress the tail of the aeroplane.

one takes a slight turn,
couple is which shows that the
nose is depressed and the tail is raised,

engine rotates or spins
on viewing from the end, then
is depressed and the tail is

eroplane takes a slight turn,
depressed and the nose is

$$N = 1800 \text{ rpm} \quad R = 250 \text{ m}, K = 0.32 \text{ m}$$

$$\frac{200}{10} = 6.94 \text{ m/s}$$

$$= 2200 \times (0.32)^2 = 225.3 \text{ kg.m}^2$$

$$\frac{2200 \times 1800}{60} = 188.5 \text{ rad/s}$$

$$= \frac{6.94}{250} = 0.0278 \text{ rad/s}$$

$225.3 \times 188.5 \times 0.0278 = 1180 \text{ N-m}$
to lower the bow (fore) and raise the
when the ship turns right.
d/s

$$\text{Wwp} = 225.3 \times 188.5 \times 0.8 = 33972 \text{ N.m}$$

of the reaction couple when the bow is
to turn the ship towards right or
starboard.

$$\omega_p = 0.1 \text{ rad/s}$$

$$C = 225.3 \times 188.5 \times 0.1 = 4246.5 \text{ Nm}$$

As the axis of spin is always parallel to the axis of precession for all positions, there is no gyroscopic effect on the ship.

$$\text{Given} \Rightarrow I_w = 2.4 \text{ kg.m}^2 \quad m = 2200 \text{ kg}$$

$$h = 0.33 \text{ m} \quad h = 0.55 \text{ m}$$

$$I_e = 1.2 \text{ kg.m}^2 \quad \omega = 1.5 \text{ m}$$

$$G = \frac{W_e}{W_w} = 3 \quad R = 80 \text{ m}$$

Reaction due to weight

$$R_w = \frac{mg}{4} = \frac{2200 \times 9.81}{4}$$

$$= 5395.5 \text{ N (upwards)}$$

Reaction due to gyroscopic couple

$$C_w = 4 I_w \frac{v^2}{4R} = \frac{4 \times 2.4 \times v^2}{0.33 \times 80}$$

$$= 0.364 v^2$$

$$C_e = I_e G \omega_w \omega_p = \frac{1.2 \times 3 \times v^2}{0.33 \times 80} = 0.136 v^2$$

$$C_h = C_w + C_e = 0.364 v^2 + 0.136 v^2 = 0.5 v^2$$

Reaction on each outer wheel,

$$R_{ho} = \frac{C_h}{2w} = \frac{0.5 v^2}{2 \times 1.5} = 0.167 v^2 \text{ (upwards)}$$

each inner wheel,
 $0.167 v^2$ (downwards)

centrifugal couple

$$h = \frac{2200 \times v^2}{80} \times 0.55 = 15.125 v^2$$

each outer wheel,

$$\frac{C_e}{2w} = \frac{15.125 v^2}{2 \times 1.5} = 5.042 v^2 \quad (\text{upwards})$$

inner wheel, $R_{ci} = 5.042 v^2$
(downward)

safe speed, the condⁿ is

$R_i + R_{ci}$

$$= (0.167 + 5.042) v^2$$

$$1035.8$$

$$32.18 \text{ m/s} \quad \text{or } v = \frac{32.18 \times 3600}{1000}$$

$$= 115.9 \text{ km/h}$$